90%

DATABASE MANAGEMENT SYSTEMS

ICT

MODULE 4

CAT 3

Candidate Name: BRIAN MAINA.

Admission Number: _____ **Date:** _____

Time: 1 hour | **Total:** 30 marks

Instructions

1. Answer ALL questions.
2. Write your name and admission number in the spaces provided above.
3. Show all working and write your answers neatly.

QUESTIONS

1. Define **normalization** and state its main purpose. (2 marks)
2. Briefly explain the three types of anomalies caused by poorly designed tables. (2 marks)
3. Differentiate between **CHAR** and **VARCHAR**, and state which data type you would use for storing prices and why. (2 marks)
4. State **two advantages** and **two disadvantages** of indexing. (2 marks)

5. Distinguish between a **PRIMARY KEY** and a **UNIQUE** constraint. (2 marks)

6. A sports club keeps the following table:

CLUB_MEMBER

MemberID	MemberName	Sports
1	Mary	Football, Netball
2	John	Chess
3	Peter	Football, Chess, Swimming

a) Explain why this table is not in 1NF. (1 mark)

b) Convert the table into 1NF and state its primary key. (3 marks)

7. Consider this table, which is in 1NF, with composite primary key {OrderID, ProductID}:

ORDER_ITEM

OrderID	ProductID	ProductName	UnitPrice	Quantity
O1	P1	Laptop	60000	2
O1	P2	Mouse	800	5
O2	P1	Laptop	60000	1

a) List the functional dependencies in the table and identify any partial dependency. (2 marks)

b) Decompose the table into 2NF. (2 marks)

c) Give **one** anomaly that the original table suffers from. (1 mark)

8. a) Write an SQL CREATE TABLE statement for **ENROLLMENT** with these attributes:
   - StudentID (integer)
   - CourseID (up to 6 characters)
   - EnrollDate (date, defaulting to the current date)
   - Marks (must be between 0 and 100)

   The primary key is {StudentID, CourseID}. StudentID references STUDENT(StudentID) and CourseID references COURSE(CourseID). (3 marks)

   b) Write a CREATE INDEX statement to index the StudentName column of the STUDENT table, and give one reason why this column is a good candidate for indexing. (2 marks)

9. Use the tables below to answer this question.

   STUDENT(StudentID, StudentName, Department)
   ENROLLMENT(StudentID, CourseID, EnrollDate, Marks)

   a) Write a SELECT query to display the names and departments of all students in the 'ICT' department, arranged alphabetically by name. (2 marks)

   b) Write a query to display the average marks for each CourseID, showing

only courses whose average is above 60. (2 marks)

c) Write a query that joins the two tables to display each student's name, CourseID and Marks. (2 marks)

(c) Primary key

(b) Unique key identifies each record in a table

(a) Does not allow null values

Unique Constraint ensures that all values in a column or group of columns are unique

Not a foreign key because it is the primary key.

(6) (a) The Table is not in 1NF because the Courses Column contains multiple values in a single cell instead of one atomic value per cell.

(b)

ID	Name	Courses
1	Mary	English
1	Mary	Drama
2	John	Dance
3	Peter	Art
3	Peter	Chess
3	Peter	Swimming

(7) Functional dependency

ProductID → ProductName

ProductID → Unit Price

(CustomerID, ProductID) → Quantity

Partial dependency

ProductID → ProductName, Unit Price

(8) To achieve Second Normal form (2NF), separate the Product information from the Order-Item information

(i) Product Table

ProductID	ProductName	Unit Price
P1	Laptop	60000
P2	mouse	800

1] Normalization. → Process of organizing database tables to reduce data redundancy & improve data integrity.

Main Purpose

- To prevent data anomalies & maintain data consistency.

2] 1) Insertion anomaly → occurs when it is impossible to insert certain data without inserting other unrelated data.

2] Update anomaly → occurs when the information is stored in several places & must be updated.

3) Deletion anomaly → occurs when deleting a record unintentionally removes other important information.

3) CHAR → Stores fixed length strings & always use extra space when the value is shorter than the defined length.

VARCHAR → Stores variable length strings, saving space according to the stored text, ideal for database-specific storage rules.

4] Advantages

1) Speeds up data retrieval & search.

2) Improves the performance of queries that filter or sort indexed columns.

Disadvantages

1) Requires additional storage space.

- Can slow down INSERT, UPDATE and DELETE operations because indexes must also be updated.

OR

(2) ORDER_ITEM Table

Order ID (PK)	Product ID (PK, FK)	Quantity
O1	P1	7
O1	P2	5
O2	P1	1

(c) Update anomaly ⇒ if the price of a product changes, every row containing that product must be updated. Otherwise, the database will contain inconsistent prices.

8]

```
CREATE TABLE ENROLLMENT (
        STUDENT ID INT,
        COURSE ID VARCHAR(6),
        ENROLLDATE DATE DEFAULT CURRENT_DATE,
        MARKS DECIMAL (5,2),

    PRIMARY (STUDENT ID, COURSE ID),

    FOREIGN KEY (STUDENTID)
        REFERENCES STUDENT (STUDENTID),

    FOREIGN KEY (COURSEID)
        REFERENCES COURSE (COURSEID),
    CHECK (MARKS BETWEEN 0 AND 100)
    );
```

(b) CREATE INDEX idx_student_name
ON STUDENT (STUDENTNAME);

05

9)

a)
```
SELECT StudentName, Department
FROM STUDENT
WHERE Department = 'ICT'
ORDER BY StudentName ASC;
```
02

b)
```
SELECT CourseID, AVG(Marks) AS Averagemarks
FROM ENROLLMENT
GROUP BY Course ID
HAVING AVG(Marks) > 60;
```
02

c)
```
SELECT
  STUDENT.StudentName,
  ENROLLMENT.CourseID,
  ENROLLMENT.Marks
FROM STUDENT
INNER JOIN ENROLLMENT
  ON STUDENT.StudentID = ENROLLMENT.StudentID;
```
02

06